

Indian Institute of Technology Jodhpur

Subject: Time Series Analysis Problem Sheet-II

Question: Answer the following questions.

1. Simulate the dataset of 300 data points from Autoregressive Moving Average series of order (2, 2) with $\phi_1 = 0.1$, $\phi_2 = 0.2$, $\theta_1 = 0.3$ and $\theta_2 = 0.3$. Plot the dataset. Also, plot ACF and PACF.
2. Consider the dataset **lake** from **itsmr** package. Determine the suitable model—AR(p) or MA(q)—and its order.
3. Consider the dataset **strikes** from **itsmr** package. Using Maximum likelihood estimator method, answer the following:
 - (a) Based on AIC measure, out of ARMA(1,1), ARMA(2,1) and ARMA(1,2) which is best fitted model.
 - (b) Based on AICc measure, out of ARMA(2,1), ARMA(2,2) and ARMA(1,2) which is best fitted model.
 - (c) Based on BIC measure, out of ARMA(2,2), ARMA(3,1) and ARMA(1,3) which is best fitted model.
4. For the best fitted model in each part of question (3), using the 5% level of significance, answer the following:
 - (a) Determine whether the residuals are iid noise or not.
 - (b) Determine whether the residuals are iid Gaussian noise or not.
5. Consider the time series having 100 observations: $X_1, X_2, X_3, \dots, X_{100}$ gave the following sample ACF: $\hat{\rho}(1) = 0.43$. Assume that the data is generated from MA(1) process. Determine the approximate 99% confidence interval for $\rho(h)$ for $h \geq 1$.
6. Consider the AR(2) process $\{X_t\}$ satisfying

$$X_t - \phi X_{t-1} - \phi^2 X_{t-2} = Z_t, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2).$$

- (a) For what values of ϕ is this a causal process?
- (b) The following sample moments were computed after observing $X_1, \dots, X_{200}$:

$$\hat{\gamma}(0) = 6.06, \quad \hat{\rho}(1) = 0.687.$$

Find estimates of ϕ and σ^2 by solving the Yule-Walker equations. (If you find more than one solution, choose the one that is causal.)

7. Consider the dataset **oil** from **fpp2** package. Using ADF, PP and KPSS test, for $\alpha = 0.05$, determine whether the series is stationary or not.

8. Consider the time series having 50 observations: $X_1, X_2, X_3, \dots, X_{50}$ generated from MA(1) process given by

$$X_t = Z_t + \theta Z_{t-1}, \quad \{Z_t\} \sim \text{IID}(0, \sigma^2).$$

The MLE estimate of θ is -0.8 . For $\alpha = 0.05$, determine whether the model has unit root or not.

9. Consider the **wine** dataset. If a seasonal component is present, eliminate it. After eliminating the seasonal component from the data, fit an appropriate ARIMA(p,d,q) model.
10. Simulate the dataset of 200 data points from SARIMA(1,0,1)(1,0,1)₁₂ with $\Phi_1 = 0.2$, $\phi_1 = 0.3$, $\Theta_1 = 0.4$ and $\theta_1 = 0.5$. Write down the model equation and plot the dataset. Also, plot ACF and PACF.
11. For each of following Seasonal ARIMA series, check for causality and invertibility.
- (a) $X_t = 0.1X_{t-12} + 0.2X_{t-24} + Z_t - 0.3Z_{t-12} - 0.3Z_{t-24}$.
 - (b) $(1 - B^5 - B^{10})(1 - 0.5B)X_t = (1 + 0.4B^5 + 0.8B^{10})(1 + 0.6B)Z_t$.
 - (c) $(1 - 0.4B^{12} - 0.6B^{24})(1 - 0.5B - 0.6B^2)X_t = (1 + 0.5B^{12} + 2B^{24})(1 + 0.8B)Z_t$.
12. Consider the dataset **elec** from **fma** package. Using Box-cox transformation, determine the value of lambda. Based on AICC and BIC, out of ARIMA(2,1,1)(1,1,1), ARIMA(2,1,2)(2,1,2) and ARIMA(0,1,1)(1,2,1) which is best fitted model.
13. For best fitted models in question (11), using 5% level of significance, determine whether the residuals are iid noise or not.
14. Consider the dataset **speech** from **astsa** package. Using ADF, PP and KPSS test, for $\alpha = 0.05$, determine whether the series is stationary or not.
15. Consider the dataset **woolryrnq** from **fma** package. Fit an appropriate SARIMA model.